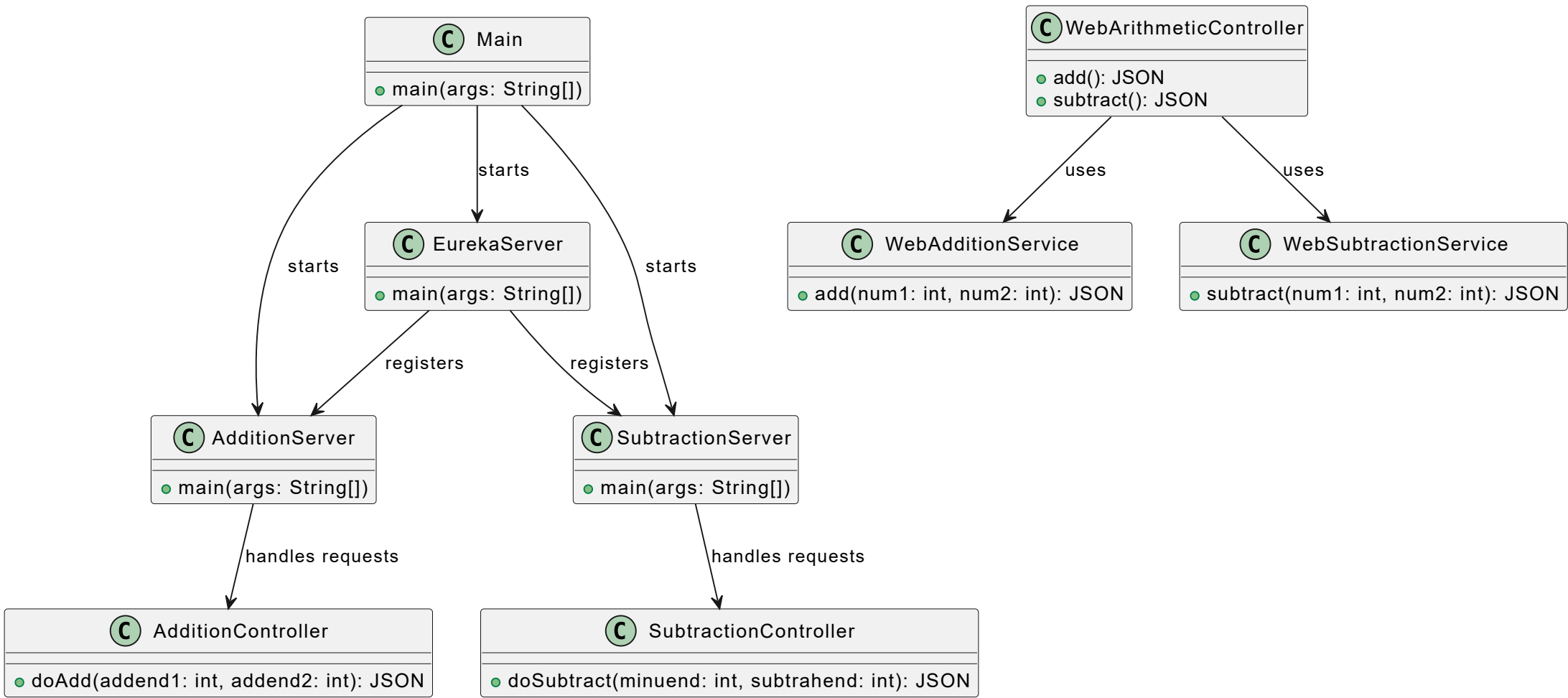


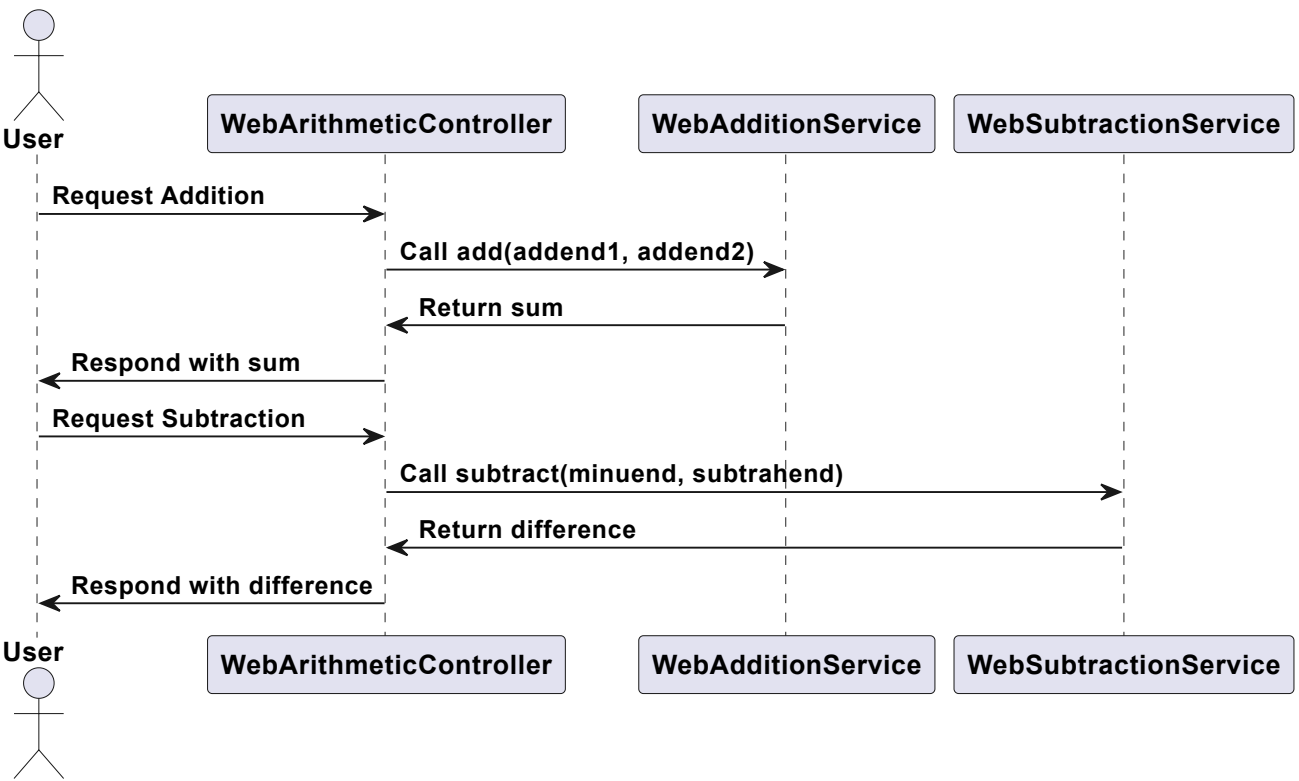
1) Class Diagram:

This class diagram represents the structure of a microservices application built with Spring Boot. It includes key classes such as Main, EurekaServer, AdditionController, SubtractionController, and their respective server classes. The diagram illustrates the relationships between these classes, showing how the Main class initiates the application, registers services with the Eureka server, and how controllers handle requests for addition and subtraction operations. Additionally, it highlights the WebArithmeticController's dependency on the WebAdditionService and WebSubtractionService for performing arithmetic operations.



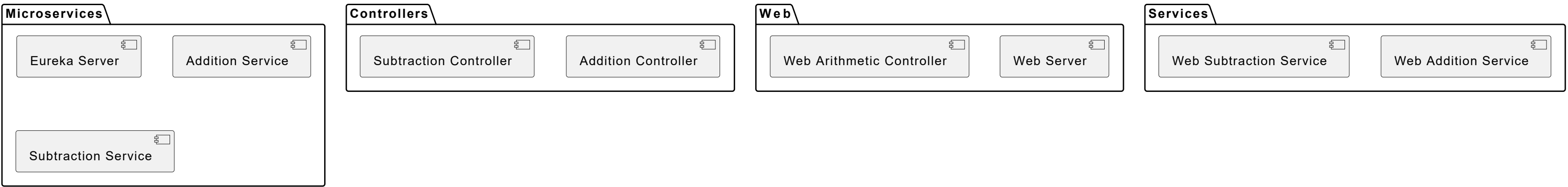
2) Sequence Diagram:

This sequence diagram illustrates the interaction between a user and the web arithmetic controller for performing addition and subtraction operations. The user initiates requests for addition and subtraction, which are handled by the WebArithmeticController. The controller calls the respective services (WebAdditionService and WebSubtractionService) to perform the calculations and returns the results back to the user.



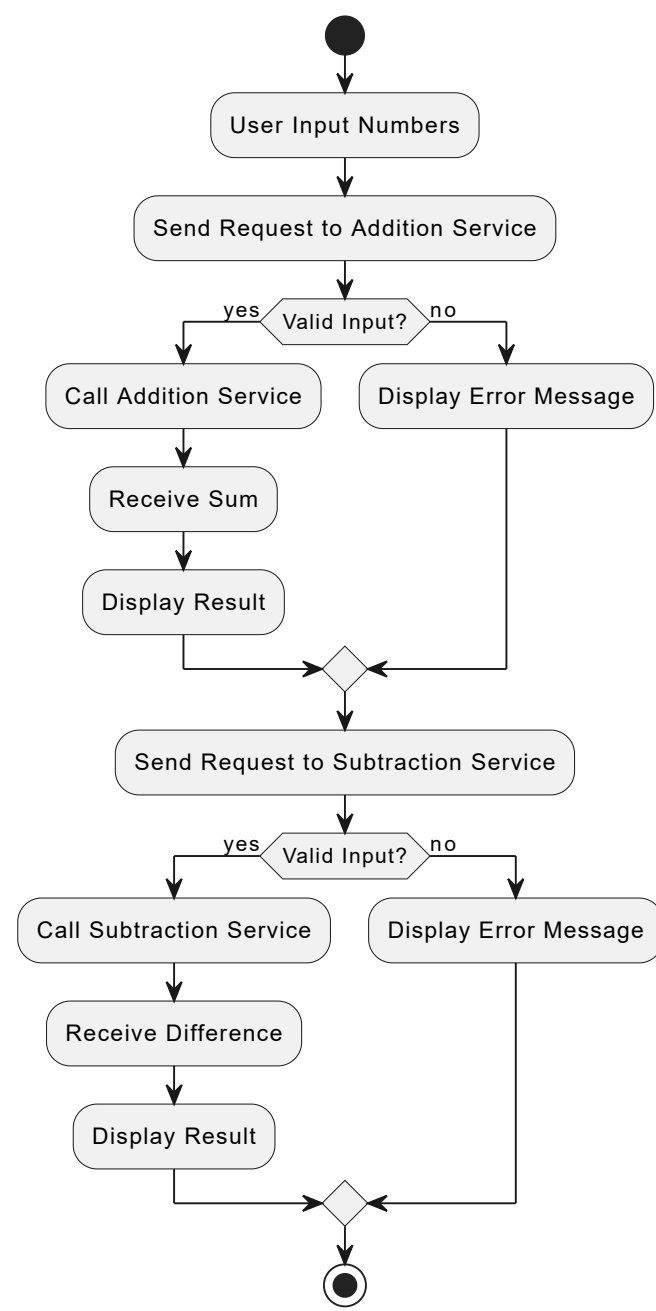
3) Logical Diagram:

This logical diagram represents the architecture of a microservices-based application built using Spring Boot. It includes the Eureka server for service discovery, the Addition and Subtraction services, and their respective controllers. The web server and web arithmetic controller facilitate user interactions, while the web addition and subtraction services handle the arithmetic operations. The diagram illustrates the modular design and communication flow between the various components.



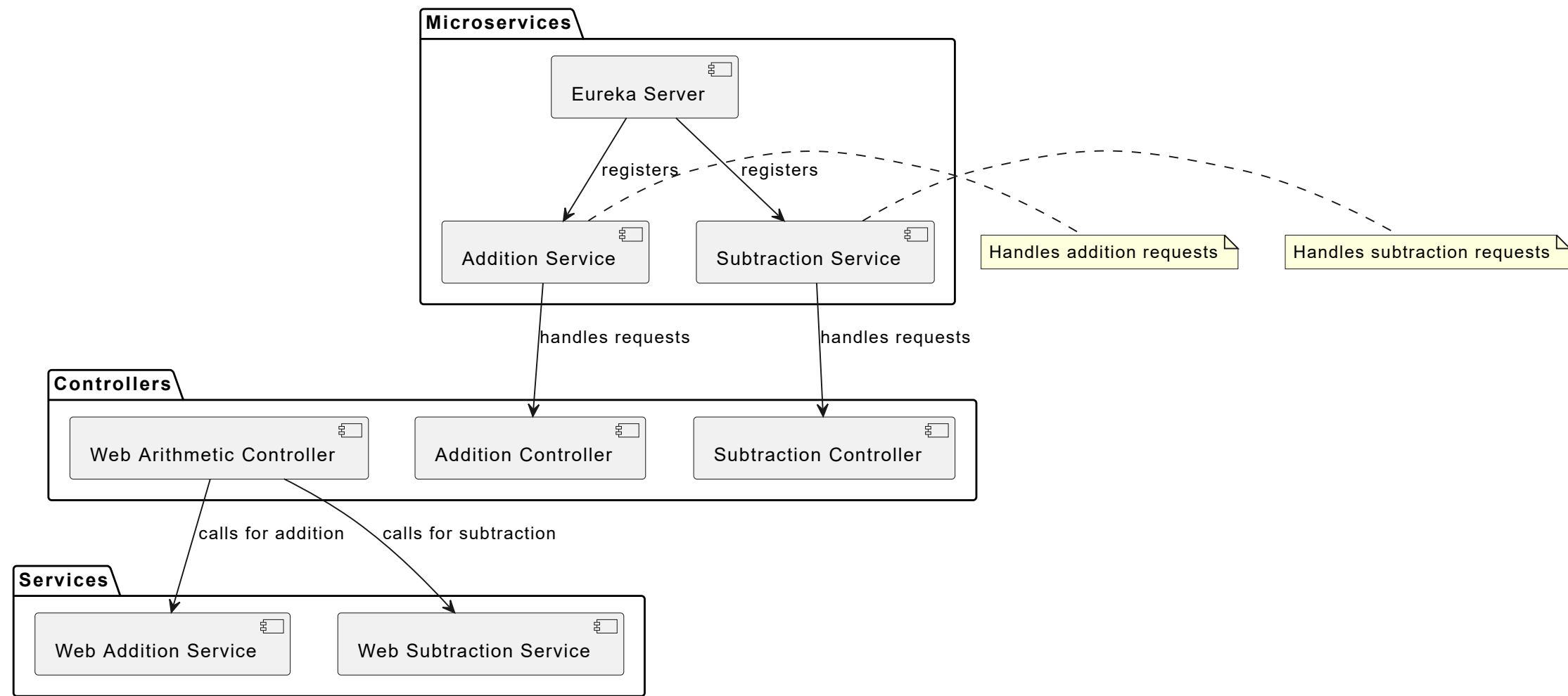
4) Activity Diagram:

This activity diagram illustrates the process of a user interacting with the microservices application. It starts with the user inputting numbers, followed by sending requests to the addition and subtraction services. The diagram includes decision points for validating input, calling the respective services, and displaying results or error messages based on the validity of the input.



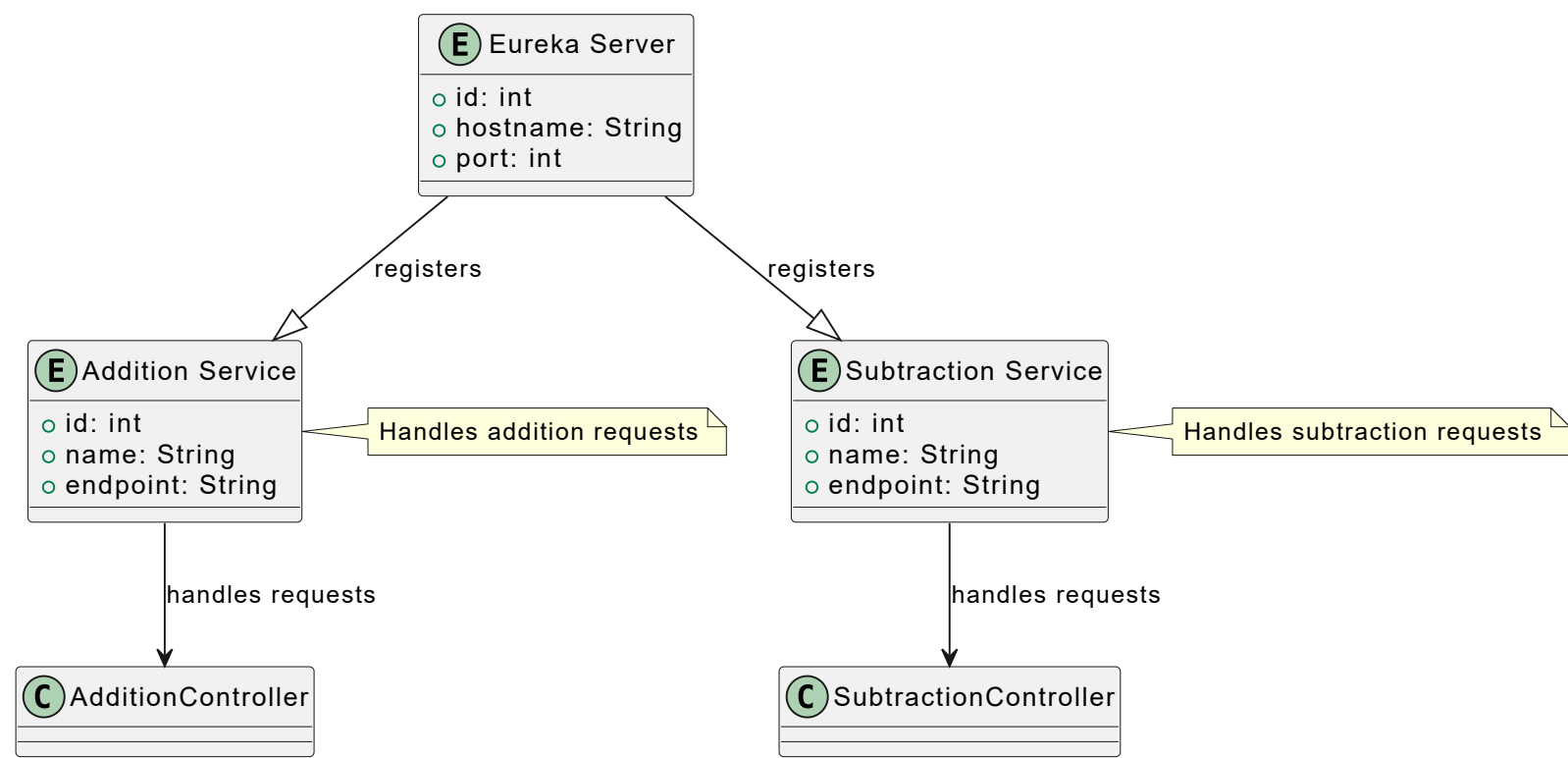
5) Component Diagram:

This component diagram illustrates the architecture of the microservices application. It shows the Eureka server facilitating service discovery for the Addition and Subtraction services. Each service has its respective controller handling incoming requests. The `WebArithmeticController` interacts with the `Web Addition` and `Subtraction` services to perform operations, demonstrating the modular design and communication between components.



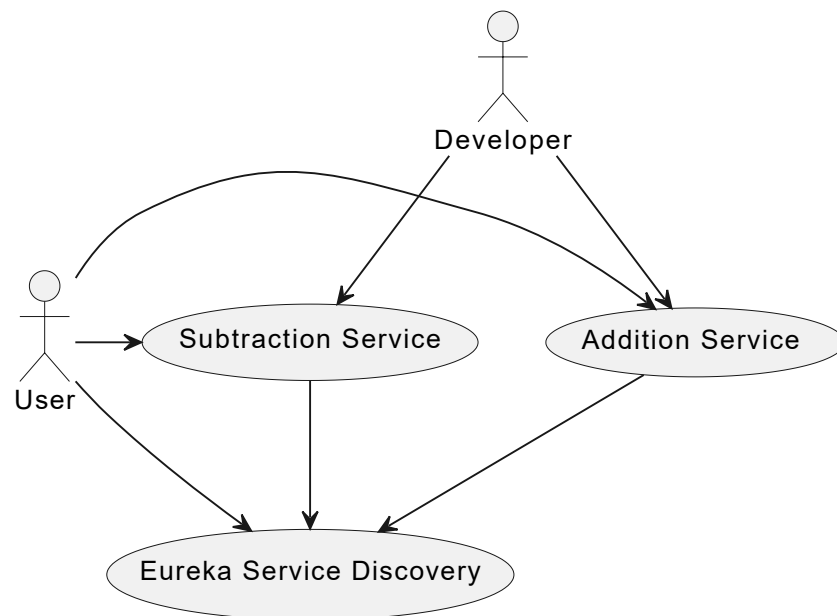
6) ER (Entity-Relationship) Diagram:

The ER diagram illustrates the relationships between the Eureka Server and the two microservices: Addition Service and Subtraction Service. The Eureka Server is responsible for registering both services, allowing them to communicate dynamically. Each service has its own controller to handle requests, with the Addition Service managing addition operations and the Subtraction Service managing subtraction operations.



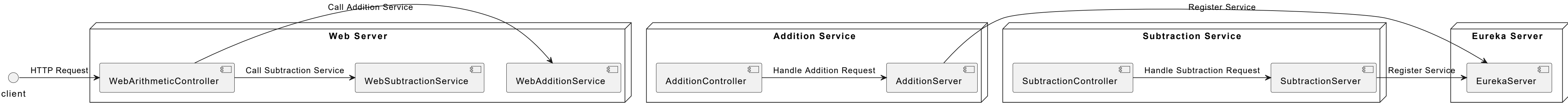
7) Use Case Diagram:

This Use Case Diagram illustrates the interactions between users, developers, and the microservices in the application. Users can access the Addition and Subtraction services, while both services register themselves with the Eureka Service Discovery for dynamic communication. Developers interact with the services for maintenance and enhancements.



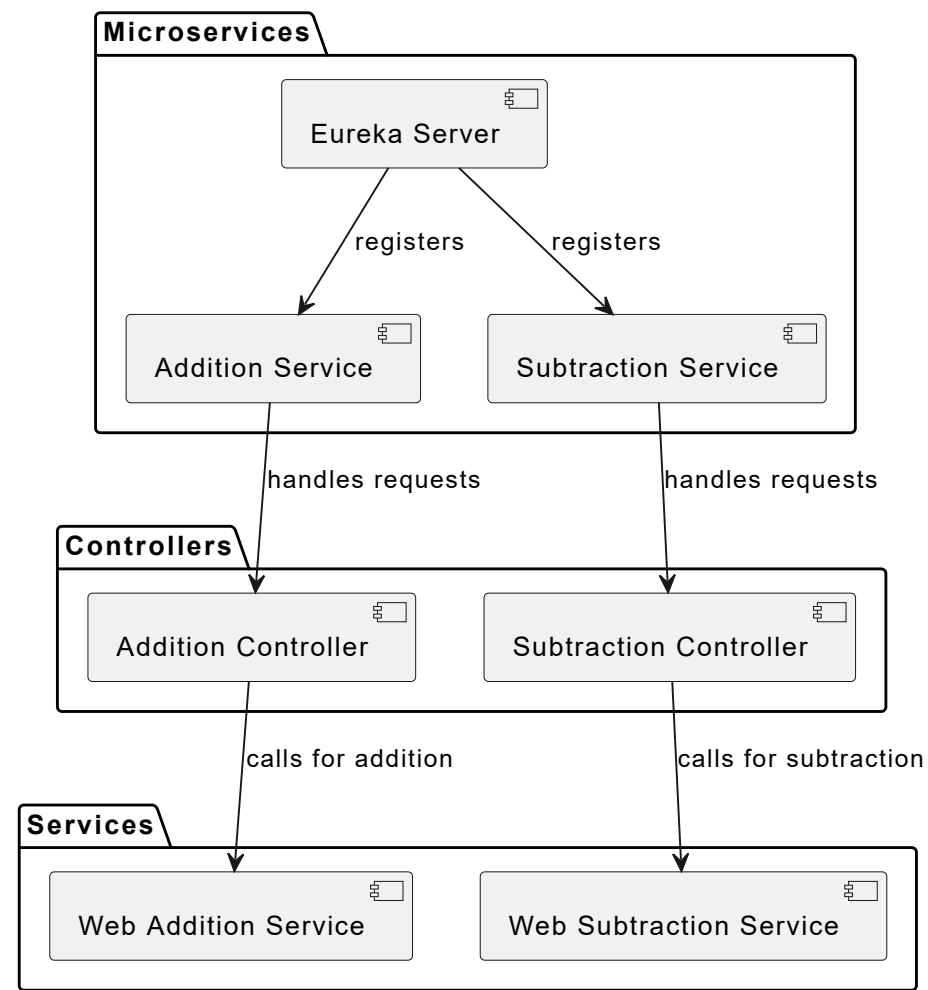
8) Deployment Diagram:

This deployment diagram illustrates the architecture of the microservices application. It includes the Eureka server for service discovery, the Addition and Subtraction services, and the Web server that interacts with clients. Each service is represented as a component within its respective node, showing how they communicate with each other and register with the Eureka server.



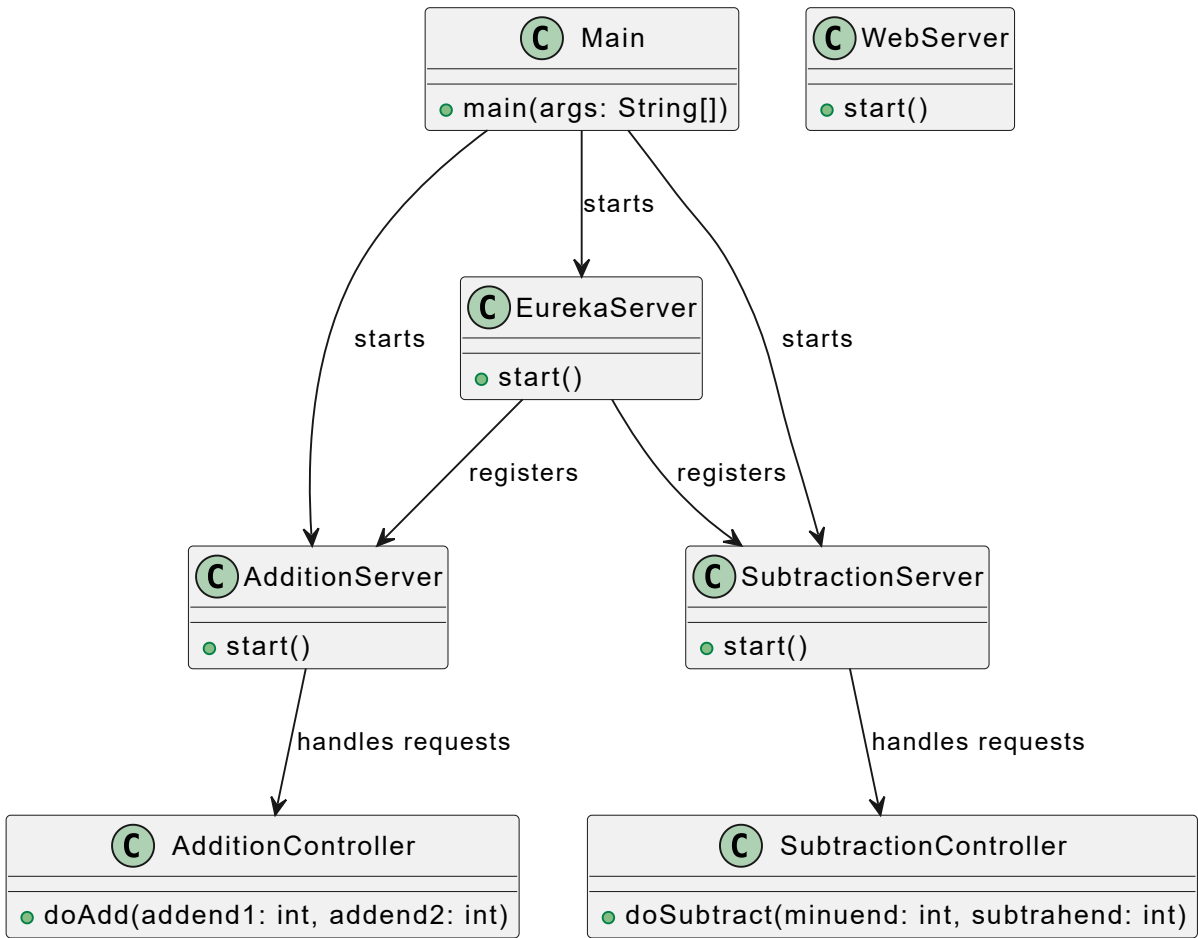
9) Composite Structure Diagram:

This Composite Structure Diagram illustrates the architecture of a microservices application. It shows the relationships between the Eureka server, which facilitates service discovery, and the Addition and Subtraction services. Each service has its respective controllers that handle incoming requests and interact with web services for performing arithmetic operations. The diagram emphasizes the modular design and interactions among components.



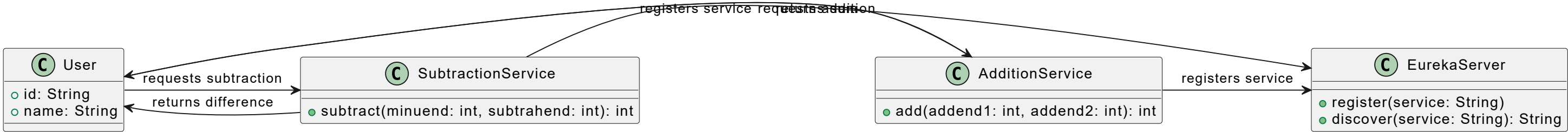
10) Object Diagram:

This object diagram illustrates the main classes of the microservices application, including the entry point (Main), the Eureka server for service discovery, and the addition and subtraction services. Each service has its own controller to handle requests. The relationships show how the Main class starts the servers, which then register with the Eureka server, and how each server interacts with its respective controller.



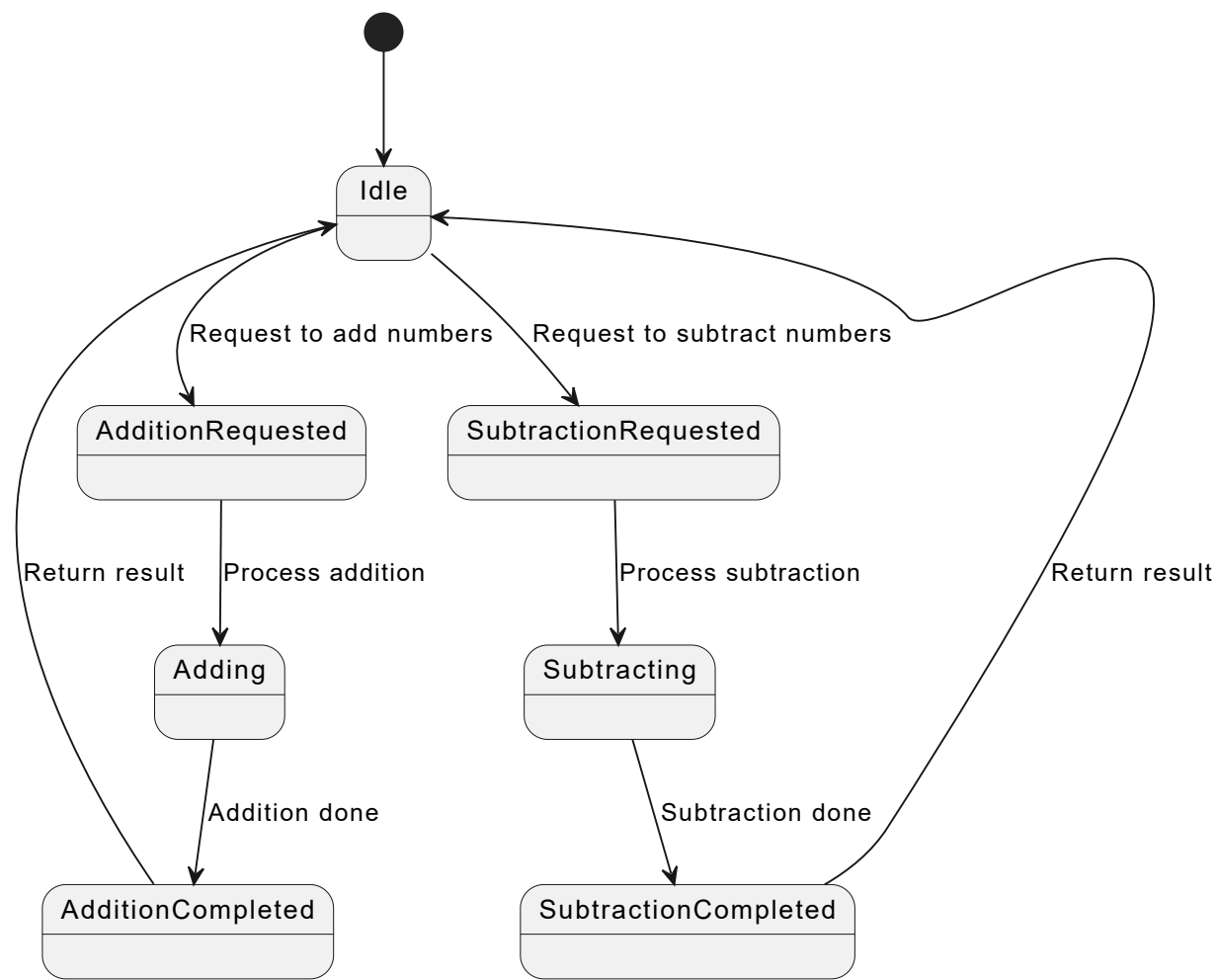
11) User Journey Map:

This User Journey Map illustrates the interactions between a User and the microservices in the application. The User can request addition and subtraction operations from their respective services. Each service registers itself with the Eureka server for service discovery, allowing dynamic communication. The services return the results of the operations back to the User.



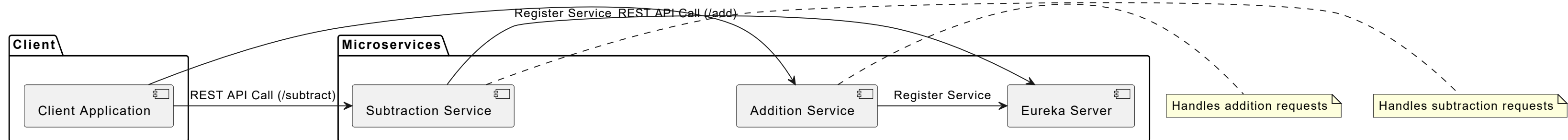
12) State Diagram:

This state diagram illustrates the states of the microservices application handling addition and subtraction requests. It starts in the 'Idle' state, transitions to 'AdditionRequested' or 'SubtractionRequested' based on user input, processes the respective operation, and returns to 'Idle' after completing the operation.



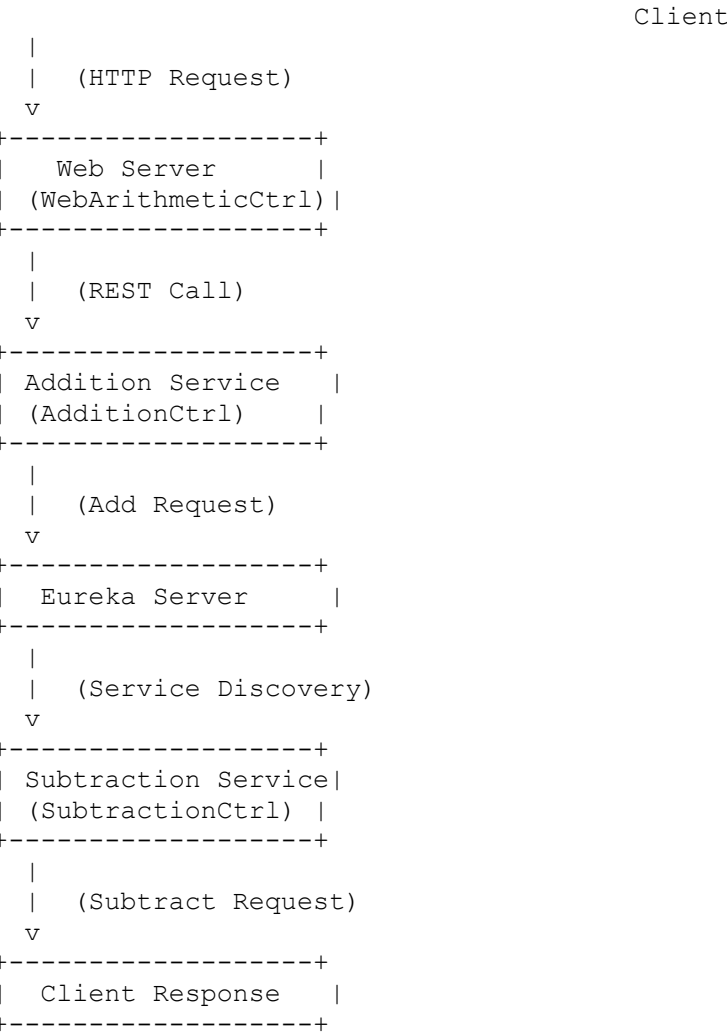
13) Architecture Diagram:

This architecture diagram illustrates a microservices-based application consisting of an Eureka server for service discovery and two microservices: Addition Service and Subtraction Service. The client application interacts with these services via REST API calls. Each service registers itself with the Eureka server, enabling dynamic service discovery and communication.



14) Workflow Diagram:

This workflow diagram illustrates the interaction between the client, web server, and microservices in the application. The client sends an HTTP request to the web server, which routes the request to the appropriate arithmetic service (either addition or subtraction) via REST calls. The services communicate with the Eureka server for service discovery, allowing them to locate each other dynamically. Finally, the results are returned to the client.



15) Communication Diagram:

This communication diagram illustrates the interactions between the user, web server, Eureka server, addition service, and subtraction service. The user initiates requests to the web server for addition and subtraction operations, which in turn communicates with the respective services to perform calculations. The web server also registers itself with the Eureka server for service discovery.

